



Vienna, Austria

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Alessandro Chessari

Education and training

- 2025-2026 **TU Wien**, Vienna, Austria.
Postdoctoral researcher in Computational Quantum Physics
- 2024-2025 **TU Wien**, Vienna, Austria.
Visiting PhD student
- 2022-2025 **CEA Grenoble**, Grenoble, France.
PhD in Theoretical Physics (Quantum Physics)
- 2021-2022 **École Normale Supérieure de Lyon**, Lyon, France.
M2 Theoretical Physics
- 2020-2021 **École Normale Supérieure de Lyon**, Lyon, France.
M1 Physics, *High Honors*
- 2019-2022 **Politecnico di Milano**, Milano, Italy.
MSc Physics Engineering, *final grade 110/110*
- 2016-2019 **Politecnico di Milano**, Milano, Italy.
BSc Physics Engineering
- 2011-2016 **Liceo Scientifico Benedetto Cairoli**, Vigevano, Italy.
Scientific High School Diploma

Fellowships

- 2024-2025 **IDEX Formation Scholarship and CEA PACE Award**, Grenoble, France.
Awarded the IDEX Formation Scholarship and CEA PACE Award, providing funding for a visiting period abroad, granted by Université Grenoble Alpes and CEA Grenoble, respectively
- 2020-2021 **IdexLyon Excellence Scholarship**, Lyon, France.
Awarded with the IdexLyon excellence scholarship, monthly scholarship for studies in Lyon, granted by École Normale Supérieure de Lyon and Université Claude Bernard de Lyon
- 2020-2022 **Double Degree Program**, Milano, Italy.
Selected for the Double Degree Program between Politecnico di Milano and École Normale Supérieure de Lyon, granted by Erasmus

Publications

- Efficient Optimization of Tensor Rings with Low-Rank Environments, (2026).
M. Jeannin, **A. Chessari**, J. V. Delft
- Many-Body Admittance Spectroscopy, (2026).
A. Chessari, M. Filippone

- Parametric longitudinal coupling of a semiconductor charge qubit and an rf resonator, *Phys. Rev. Applied* 23, 034067 (2025).

V. Champain, S. Zihlmann, **A. Chessari**, B. Bertrand, H. Niebojewski, E. Dumur, X. Jehl, V. Schmitt, B. Brun, C. Winkelmann, Y.-M. Niquet, M. Filippone, S. D. Franceschi, R. Maurand

- Unifying Floquet theory of longitudinal and dispersive readout, *Phys. Rev. Lett.* 134, 037003 (2025).

A. Chessari, E. A. Rodríguez-Mena, J. C. Abadillo-Uriel, V. Champain, S. Zihlmann, R. Maurand, Y.-M. Niquet, and M. Filippone

- Role of electron polarization in nuclear spin diffusion, *PRB* 107, 224429 (2023).

A. Chessari, S. F. Cousin, S. Jannin and Q. Stern

Teaching Experiences

September - **Exercise session for "Electrical phenomena and transport"**, *Université*
December *Grenoble-Alpes*, Grenoble, France.
2024

September - **Exercise session for "Electrical phenomena and transport"**, *Université*
December *Grenoble-Alpes*, Grenoble, France.
2023

February - **Exercise session for "Introduction to Linear Algebra"**, *Université Grenoble-*
June 2023 *Alpes*, Grenoble, France.

Research Experiences

December **Postdoctoral researcher**, *TU Wien*, Vienna, Austria.
2025 - Present Postdoc researcher exploring quantum inspired tensor networks algorithms. Advisors Dr. Anna Kauch and Dr. Markus Wallerberger.

December **Visiting PhD student**, *TU Wien*, Vienna, Austria.
2024 - June Visiting period at TU Wien investigating the applicability of Quantics Tensor Train in one
2025 dimensional spin chains, advisors Dr. Anna Kauch and Dr. Markus Wallerberger.

October 2022 - **PhD in Theoretical Physics**, *CEA*, Grenoble, France.
October 2025 From single qubit to many-body readout in Circuit Quantum Electrodynamics platforms. Advisors Dr. Michele Filippone and Dr. Y.-M. Niquet.

March-July **Master Student**, *CEA*, Grenoble, France.
2022 Admittance calculation for a Quantum Many-Body dissipative system.
Master Thesis project, theoretical research in Quantum dissipative Many-Body systems, under the supervision of Research Engineer Michele Filippone.

May-July **Master Student**, *CRMN Lab*, ENS de Lyon, Lyon, France.
2021 Spin Diffusion in Solid State.
Theoretical research activity in the RMN field, in particular focusing on the polarisation transfer between nuclear spins into paramagnetic solids.
This internship was supervised by Dr. Quentin Stern.

June-July **Bachelor Student**, *Semi-Spin Lab*, Politecnico di Milano, Milano, Italy.
2019 Inverse Spin-Hall Effect in sample of Pt/Cu/Ge and Au/Cu/Ge. The aim of this short period of research in the SemiSpin Lab at Politecnico di Milano was to detect spin current generated by circularly polarized light in germanium via the Inverse Spin-Hall effect in platinum and gold.
This internship was supervised by Dr. Federico Bottegoni and Prof. Marco Finazzi.

Schools and Conferences

July 2026 **International workshop on tensor cross interpolation and other algorithms for learning tensor networks**, *Grenoble, France*.
Tensor networks workshop in Grenoble.

- October 2025 **International workshop on tensor cross interpolation and other algorithms for learning tensor networks**, *Grenoble, France*.
Tensor networks workshop in Grenoble.
- August 2024 **SiQEW2024**, *Davos, Switzerland*.
Silicon Quantum Electronics Workshop.
- January - **Trimester "Quantum many-body systems out-of-equilibrium"**, *Paris, France*.
March 2024 Thematic trimester program at Institut Henri Poincaré.
- 26-3 December **Introductory school to Quantum many-body systems out of equilibrium**,
2023 *Cargese, France*.
Introductory school part of the thematic trimester program "Quantum many-body systems out-of-equilibrium".
- 28-3 December **GDR Quantum Mesoscopic Physics**, *Aussois, France*.
2022 Conference about Mesoscopic Physics at CNRS Center Paul Langevin in Aussois.
- 16-18 **GDR IQFA 13**, *Paris, France*.
November Conference about Quantum Engineering, Fundamental Aspects to Applications at C2N
2022 Center in Saclay.
- 10-21 October **Doctoral Training "Frontiers of Condensed Matter"**, *Les Houches, France*.
2022 Selected for the Summer School in Les Houches.
- 9-12 February **Winter School on Ultracold Quantum Gases**, *Innsbruck, Austria*.
2022 Selected for the Winter School held in Innsbruck. Twelves lectures given by experts from Innsbruck, Heidelberg, Wien and Trento covering the basic concepts of laser cooling and trapping, Bose-Einstein condensation, ultracold Fermi gases, ultracold molecules, strongly correlated quantum gases, and optical lattices.
- 5-9 September **Hyp21 Conference**, *International Hyperpolarization Conference*, Lyon, France.
2021 Participation at the Hyp21 Conference as a student member of Sami Jannin équipe. Citation in the talk given by Quentin Stern.

Technical Skills

Advanced C, C++, JULIA, PYTHON, L^AT_EX, MATLAB

Languages

Italian **Mothertongue**
Romanian **Second Language**
English **Fluent**
French **Fluent**
German **B2 Level**